The “Difficulties” in Fermat’s Original Discourse on the Indecomposability of Powers Greater Than a Square: A Retrospect

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Abstract

We present a reconstruction of Pierre Fermat’s manuscript based on the two-power decomposition constraint. A functional scaling factor $o(n) > 1$ is defined, where $o(n)$ is used only as a notation for a parameter depending on n, and not in the asymptotic little-o sense. This factor is defined in such a way that any hypothetical natural solution of the equation

$$x^n + y^n = z^n$$

with $n > 2$ entails the identity

$$o(n)^n = 2n.$$

Since Fermat’s Last Theorem concerns the decomposition of a power into exactly two powers, the algebraic roots-of-unity filter for symmetric polynomials, combined with the boundary normalization of the odd binomial sum ($\lim_{j \to 1^+} q(j) = 2$), motivates and fixes, under the global binary-normalization hypothesis, the uniform choice $o(n) = 2$, representing this binary split. The formal development in Coq then verifies that the resulting relation

$$2^n = 2n,$$

which represents the binary case of the general transcendental comparison $k^n = kn$, holds only for the exponents $n \in \{1, 2\}$. Thus, any supposed counterexample with $n > 2$ leads to a contradiction within the binary-normalization hypothesis.

The decomposition condition is kept as an explicit hypothesis over $\mathbb{N}$, while bridge lemmas connect the real equality

$$o(n)^n = 2n$$

with the integer comparison between 2^n and $2n$ under the condition $o(n) = 2$. The classical parametrization

$$(z, x) = (m^n + p^n,\ m^n - p^n)$$

and the parity identities are included exclusively as motivation and consistency checks, and are not required for the final conditional contradiction.

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1 Introduction

The present work is the result of an attempted reconstruction of Fermat's original discourse along with an explanation of why he might have not written it down. The author had performed it within a two-time period of time– between 1990 and 1993 – trying to prove the theorem and the final revision in 2017-2025.[1] When completed, it did look like a proof of Fermat's epoch, as it only involved the knowledge and techniques available and utilised by Fermat's contemporary and pre-Fermat mathematical world.

Not to overburden this text with details of a real historical study, let us briefly recall the history of the conjecture. Around 1637, Fermat wrote his Last Theorem in the margin of his copy of the Arithmetica next to Diophantus' sum-of-squares problem [7]:

Cubum autem in duos cubos, aut quadratoquadratum in duos quadratoquadra-tos & generaliter nullam in infinitum ultra quadratum potestatem in duos eiusdem nominis fas est dividere cuius rei demonstrationem mirabilem sane detexi. Hanc marginis exiguitas non caperet.

Attempts to prove this conjecture employed a diverse range of methods. Early attempts, including Fermat's own proof for the case $n = 4$, utilized the **method of infinite descent**. A significant step forward was Leonhard Euler's proof for the case $n = 3$, which involved the use of **Gaussian integers** [13]. Subsequent efforts included approaches based on the work of Sophie Germain and other techniques, before Andrew Wiles finally presented a complete proof in 1995, drawing upon deep results from **the theory of modular forms and elliptic curves** [1–6].

Modern methods, such as the theory of modular forms, deal with **the transformation properties of specific curves over particular types of spaces** (e.g., rational numbers), highlighting the **stability of elliptic curves with respect to modular transformations**. While the contemporary formalizations of algebraic curves, spaces, transformations, and groups used in these methods were not present in Fermat's time, mathematicians of that era possessed their own approaches and intuitive understandings for studying the properties of natural numbers (and primes). (More on this can be found in the Conclusions section of this paper.)

Theorem 1 (Conditional on the two-power decomposition constraint)**.** *Let*

$$o(n) = \frac{p^n q}{l}$$

be the functional scaling factor associated with a hypothetical positive natural solution to the Fermat equation. Assume that for every integer $n > 2$ and all $x, y, z \in \mathbb{N}_{>0}$,

$$x^n + y^n = z^n \implies o(n)^n = 2n.$$

Assume further the global binary-normalization hypothesis: the binary symmetric structure of Fermat's equation, where the decomposition is into exactly $k = 2$ powers, together with the algebraic roots-of-unity filter for symmetric polynomials and the continuous boundary normalization of the odd binomial sum,

$$\lim_{j \to 1^+} q(j) = 2,$$

selects the uniform global scaling value

$$o(n) = 2.$$

Then the Fermat equation

$$x^n + y^n = z^n$$

has no positive natural solutions for any $n > 2$.

[1] *The work was mainly performed on the scholarship of the NRNU MEPhI student in* $1990 - 93$, *final revision in* $2017 - 2025$.

Proof sketch. Given a putative positive natural solution at a fixed exponent $n > 2$, the two-power decomposition constraint yields the scaling relation

$$o(n)^n = 2n.$$

The local binomial scaling parameter $q(j)$ is not identified pointwise with the global scaling factor $o(n)$. Rather, the global binary-normalization hypothesis uses the symmetric boundary value

$$\lim_{j \to 1^+} q(j) = 2$$

to select the normalized global value

$$o(n) = 2.$$

Substituting this normalization into the scaling relation gives

$$2^n = 2n.$$

This is the binary case $k = 2$ of the general exponential-linear comparison

$$k^n = kn.$$

For $k \geq 3$, the relation $k^n = kn$ has only the trivial positive integer solution $n = 1$. The binary case $k = 2$ is exceptional: it has exactly two positive integer solutions,

$$n \in \{1, 2\}.$$

Equivalently,

$$2^n > 2n$$

for every integer $n > 2$. Hence a putative solution with $n > 2$ contradicts the normalized scaling relation. A machine-checked formalization of this growth comparison is provided in the accompanying Coq file `GlobalNormalization.v`.

2 Possible Proof

The statement of the theorem is rather straightforward and as follows:
Neither a cube for two cubes, nor a biquadrate or two biquadrates, and generally no power greater than two can be decomposed into two powers of the same grade. In other words, the equation

$$x^n + y^n = z^n$$

has no solutions in natural numbers if n is an integer greater than 2.
Therefore, first

1). Let's write down the theorem

$$x^n + y^n = z^n \tag{2.1}$$

2). Let's rewrite (2.1)

$$z^n - x^n = y^n \tag{2.2}$$

3). Let's perform the transformation of (2.2) to

$$\big(m^n + p^n\big)^n - \big(m^n - p^n\big)^n = y^n, \tag{2.3}$$

where $n \in \mathbb{N}$, m, p are arbitrary numbers (not necessarily integers; signs arbitrary), and $z, x \in \mathbb{N}$ satisfy

$$\begin{cases} m^n + p^n = z, \\ m^n - p^n = x. \end{cases} \tag{2.4}$$

Raising both identities to the n-th power yields

$$\begin{cases} (m^n + p^n)^n = z^n, \\ (m^n - p^n)^n = x^n. \end{cases} \tag{2.5}$$

Thus (2.4) implies (2.5).

Solving (2.4). Adding and subtracting the equations gives

$$2m^n = z + x, \qquad 2p^n = z - x,$$

so formally

$$m = \sqrt[n]{\frac{z+x}{2}}, \qquad p = \sqrt[n]{\frac{z-x}{2}}.$$

Domain note. Over $\mathbb{R}$: for odd n the roots are unique; for even n we need $(z \pm x)/2 \geq 0$ and there is a sign choice $m = \pm\big((z+x)/2\big)^{1/n}$, $p = \pm\big((z-x)/2\big)^{1/n}$. Over $\mathbb{C}$: there are n branches for the n-th root; once a branch is fixed, the reconstruction is consistent.

Equivalence of (2.4)–(2.5). Forward (2.4) $\Rightarrow$ (2.5). Immediate by raising to the n-th power.

Reverse (2.5) $\Rightarrow$ (2.4). Assume $z, x \in \mathbb{N}$ and

$$z^n = (m^n + p^n)^n, \qquad x^n = (m^n - p^n)^n.$$

Taking n-th roots in the same domain as $m^n \pm p^n$ (principal root over $\mathbb{R}_{\geq 0}$, or a fixed branch over $\mathbb{C}$) gives

$$z = m^n + p^n, \qquad x = m^n - p^n,$$

i.e. (2.4). For even n one fixes consistent signs as above. Hence, (2.4) and (2.5) are equivalent in the chosen number domain once the root/branch convention is fixed.

Proposition 1 (Integer case: necessary & sufficient conditions). *If, <u>in addition</u>, one requires $m, p \in \mathbb{Z}$, then necessarily*

$$z \pm x \text{ are even}, \qquad \frac{z+x}{2} = m^n, \qquad \frac{z-x}{2} = p^n.$$

Conversely, if $z \pm x$ are even and both halves are perfect n-th powers in $\mathbb{Z}$, then the reconstructed m, p are integers (up to signs for even n).

Remark 1. *This separates the general real/complex reconstruction (no parity obstruction) from the integer reconstruction, where parity and perfect-power constraints are essential.*

Example (real domain). For $n = 2$, $m = 3$, $p = 2$:

$$z = 3^2 + 2^2 = 13, \qquad x = 3^2 - 2^2 = 5,$$

hence

$$z^2 = 169, \qquad x^2 = 25,$$

and reversing via $(z \pm x)/2$ returns $m = 3$, $p = 2$.

Counterexample (integer case). For $n = 3$, $z = 2$, $x = 1$ we have $(z + x)/2 = 3/2$, which is not a perfect cube in $\mathbb{Z}$; hence no integer m, p satisfy (2.4), although real m, p exist.

Modular remark. The present work does not construct a proof by means of modular arithmetic and does not claim the absence of solutions to the corresponding congruences over finite fields. The scheme considered here concerns equalities over the natural, integer, rational, and real numbers, not equations over a finite field $\mathbb{F}_q$.

In particular, from an integer equality

$$x^n + y^n = z^n$$

one indeed obtains the congruence modulo any q:

$$x^n + y^n \equiv z^n \pmod{q}.$$

However, the converse implication is not valid in general: from the congruence

$$x^n + y^n \equiv z^n \pmod{q}$$

one cannot infer the equality

$$x^n + y^n = z^n$$

over the integers.

In other words, a modular congruence only means that there exists an integer t such that

$$z^n = x^n + y^n + qt.$$

An integer solution of Fermat's equation corresponds to the special case

$$t = 0.$$

Therefore, the existence of solutions to the congruences

$$x^n + y^n \equiv z^n \pmod{q}$$

for $n > 2$ is not a counterexample to Fermat's Last Theorem and does not disprove the real parametrization used in the present work.

If one specifically passes to a modulus q, then the parameters must be specified separately at the modular level. For example, instead of the real equalities

$$z = m^n + p^n, \qquad x = m^n - p^n$$

one should separately fix the residues

$$m^n \equiv A \pmod{q}, \qquad p^n \equiv B \pmod{q},$$

after which the corresponding modular relations take the form

$$z \equiv A + B \pmod{q}, \qquad x \equiv A - B \pmod{q}.$$

Thus, the real parametrization and modular arithmetic belong to different levels of description. In this article, the first level is used: equalities over $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$, rather than a separate theory of solutions over finite fields.

4). Let's ask ourselves the question: what is the number y? Is it positive integer or not? If it is not positive integer, then under what conditions will it be natural number? Does its naturalness depend on the degree of n?

From (2.3) we have the difference:

$$y^n = z^n - x^n = (m^n + p^n)^n - (m^n - p^n)^n = \tag{2.6}$$

that can be expanded or decomposed into a sum according to Newton's binomial [8, 9]:

$$\begin{aligned}&= [(m^n)^n + C_n^1(m^n)^{n-1}p^n + C_n^2(m^n)^{n-2}(p^n)^2 + \cdots + C_n^{n-1}m^n(p^n)^{n-1} + (p^n)^n] - \\ &-[(m^n)^n - C_n^1(m^n)^{n-1}p^n + C_n^2(m^n)^{n-2}(p^n)^2 \pm \cdots \pm C_n^{n-1}m^n(p^n)^{n-1} \pm (p^n)^n] = \\ &= 2C_n^1(m^n)^{n-1}p^n + 2C_n^3(m^n)^{n-3}(p^n)^3 + \cdots + 2C_n^k(m^n)^{n-k}(p^n)^k + \{2C_n^n(p^n)^n\} = \\ &= 2\textstyle\sum_{i=0}^k C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{(2i+1)}\end{aligned}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even.

Rewrite then (2.6) as

$$z^n = x^n + y^n,$$

$$\text{where } x,\ y,\ z \text{ are} \begin{cases} z = m^n + p^n \\ x = m^n - p^n \\ y = \sqrt[n]{2}\left[\sum_{i=0}^k C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{(2i+1)}\right]^{1/n} \end{cases}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even;

we know that in common case for $n = 2$ we have

$$\text{where } x,\ y,\ z \text{ are given by} \begin{cases} z = r \cdot (m^2 + p^2), \\ x = r \cdot (m^2 - p^2), \\ y = r \cdot 2mp. \end{cases} \tag{2.7}$$

but in our case, we omit the factor r – is some positive ($r > 0$) integer constant since it is reduced in our calculations. Therefore, we made the transformation (2.3) in order to take into account this case (n=2), known since the time of Pythagoras, since the general case should include a particular solution as a subset.

5). scrutinise now the y:

$$y = \sqrt[n]{2}\left[\sum_{i=0}^k C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{(2i+1)}\right]^{1/n}. \tag{2.8}$$

In order for the y to maybe a positive integer, $\sqrt[n]{2}$ must leave, since for $n > 1$ $\sqrt[n]{2}$ is an irrational number (see Appendix A). It is thus necessary that the expression

$$\left[\sum_{i=0}^{k} C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{(2i+1)}\right]^{1/n} \tag{2.9}$$

contain some common factor that destroys the radical expression $\sqrt[n]{2}$, let's find out what it is. Otherwise, y is not a positive integer due to the presence of $\sqrt[n]{2}$. Consider now what largest divisor this sum may contain and what it is equal to:

$$\begin{aligned}
&\left[\sum_{i=0}^{k} C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{(2i+1)}\right] = \\
&= C_n^1(m^n)^{n-1}p^n + C_n^3(m^n)^{n-3}(p^n)^3 + \cdots + C_n^k(m^n)^{n-k}(p^n)^k + \{C_n^n(p^n)^n\} = \\
&= n(m^n)^{n-1}p^n + \tfrac{n(n-1)(n-2)}{3!}(m^n)^{n-3}p^3 + \cdots + \\
&+ \tfrac{n(n-1)\ldots(n-k+1)}{k!}(m^n)^{n-k}(p^n)^k + \left\{\tfrac{n(n-1)\ldots 2\cdot 1}{n!}(p^n)^n\right\} = \\
&= n\cdot[(m^n)^{n-1}p^n + \tfrac{(n-1)(n-2)}{3!}(m^n)^{n-3}p^3 + \ldots + \\
&+ \tfrac{(n-1)\ldots(n-k+1)}{k!}(m^n)^{n-k}(p^n)^k + \left\{\tfrac{(n-1)\ldots 2\cdot 1}{n!}(p^n)^n\right\}] = n\cdot l^n
\end{aligned} \tag{2.10}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even,

Hence the conclusion follows: the expression admits a common factor n in the real algebraic factorization. Here l denotes a scaling constant about which nothing is assumed a priori; in the general case it may be real, and in special arithmetic cases it may be integer. It represents the remaining part of the radical after extracting the common factor n.

Since each term in the expression can be written with a common factor n, the whole sum can be factored over the real domain in the form

$$\sum_{i=0}^{k} C_n^{(2i+1)}(m^n)^{n-(2i+1)}(p^n)^{2i+1} = n\,l^n.$$

Here l is a real scaling parameter defined by this factorization. Thus, this step should be understood as extracting a common real factor n, not as claiming integer divisibility of the whole expression by n.

To show the uniqueness of the decomposition of expression (2.10), we can use the following considerations:

(a) Degree n is a fixed positive integer.

(b) Exponentiation is an unambiguous operation; for any number, there is a unique value for the degree.

(c) Addition and subtraction of real numbers are commutative operations, and the result does not depend on the order of actions.

Based on these properties, it can be argued that for fixed m, p and n there is a single result of calculating the expression (2.10). Rearranging the members will not affect the final answer.

We can see from (2.8 - 2.10) (where l is some constant (which will be it for fixed m, p, n), which we don't know anything about yet)

$$y = \sqrt[n]{2}\cdot n\cdot l \tag{2.11}$$

As a result, from (2.6)–(2.11) we get

$$z^n - x^n = \left(m^n + p^n\right)^n - \left(m^n - p^n\right)^n = y^n. \tag{2.12}$$

Substituting (2.11) into (2.12), we have

$$\big(m^n + p^n\big)^n - \big(m^n - p^n\big)^n \;=\; 2 \cdot n\, l^n. \tag{2.13}$$

Let us set $m = j\,p$ with an arbitrary $j > 1$. Then

$$\big(p^n(j^n + 1)\big)^n - \big(p^n(j^n - 1)\big)^n \;=\; 2 \cdot n\, l^n \quad\Longrightarrow\quad \big(p^n\big)^n\big(\,(j^n + 1)^n - (j^n - 1)^n\big) \;=\; 2 \cdot n\, l^n.$$

The difference $(j^n + 1)^n - (j^n - 1)^n$ by the binomial theorem reduces to odd indices:

$$(j^n + 1)^n - (j^n - 1)^n = \sum_{k=0}^{n} \binom{n}{k} (j^n)^{\,n-k}\big[1 - (-1)^k\big] = 2 \sum_{\substack{1 \le k \le n \\ k \text{ is odd}}} \binom{n}{k} (j^n)^{\,n-k}.$$

Let us denote

$$q^n \;:=\; 2 \sum_{\substack{1 \le k \le n \\ k \text{ is odd}}} \binom{n}{k} (j^n)^{\,n-k}.$$

Then

$$\big(p^n\big)^n q^n \;=\; 2 \cdot n\, l^n \quad\Longrightarrow\quad \frac{\big(p^n\big)^n q^n}{l^n} = 2 \cdot n,$$

that is

$$\left(\frac{p^n q}{l}\right)^n = 2 \cdot n. \tag{2.14}$$

The two-power decomposition constraint and the function $o(n)$. Equality (2.14) shows that for any hypothetical integer solution $x^n + y^n = z^n$, the value $\frac{p^n q}{l}$ is the n-th root of $2n$. Let us define the scaling function $o(n)$ representing this value:

$$o(n) \;:=\; \tfrac{p^n q}{l} = (2 \cdot n)^{1/n}.$$

The scaling factor of 2 in the expression $2n$ is not merely coincidental, but is a fundamental algebraic consequence of the number of terms involved in the decomposition. Under the general theory of symmetric polynomials, when a power is decomposed into k symmetric variables, the modular roots-of-unity filter (or discrete Fourier transform orthogonality) isolates terms with a scaling factor precisely equal to k. For Fermat's Last Theorem, which restricts the decomposition specifically to $k = 2$ powers ($z^n = x^n + y^n$), the filter reduces to the binary roots of unity ± 1. The resulting binomial expansion of the difference of these two symmetric terms cancels out the even-indexed components, leaving the remaining odd-indexed ones scaled exactly by 2.

This algebraic choice of the base is further illuminated by an elegant combinatorial property of the binomial coefficients on the boundary of the parameter space. Recall the well-known identity for the sum of odd binomial coefficients (Corollary 1, Equation 3.2):

$$2\sum_{i=0}^{k} \binom{n}{2i+1} = 2^n.$$

While the non-trivial domain of the parameterized scaling factor $q(j)$ strictly requires $j > 1$ (since $j = 1$ corresponds to the degenerate case $x = m^n - p^n = 0$), we can analyze the limiting behavior of this local scaling function at the boundary of the parameter space. As $j \to 1^+$, the continuous extension of the local scaling function $q(j)$ to the symmetric boundary yields:

$$\lim_{j \to 1^+} q(j)^n \;=\; 2 \sum_{\substack{1 \le k \le n \\ k \text{ is odd}}} \binom{n}{k} (1)^{n-k} \;=\; 2\sum_{i=0}^{k} \binom{n}{2i+1} \;=\; 2^n \quad\Longleftrightarrow\quad \lim_{j \to 1^+} q(j) = 2.$$

Within the active domain $j > 1$, the local parameter $q(j)$ varies with j and is strictly greater than 2 for all $n > 1$ (since the terms $(j^n)^{n-k} > 1$). However, its symmetric boundary normalization yields the limiting value 2.

Local scale versus global normalization. It is important to emphasize that the local scaling function $q(j)$ is not identified pointwise with the global scaling factor $o(n)$. For $j > 1$, the quantity $q(j)$ remains a local parameter depending on the ratio

$$j = \frac{m}{p},$$

and therefore it generally varies inside the non-degenerate domain of the parametrization. In particular, the equality

$$q(j) = 2$$

is not asserted for all $j > 1$.

Only the symmetric boundary value

$$\lim_{j \to 1+} q(j) = 2$$

is used as a normalization value for the global binary scale. Thus, the passage from the local function $q(j)$ to the uniform parameter $o(n)$ is a normalization step, not a pointwise identification:

$$q(j) \not\equiv o(n).$$

Rather, the global binary-normalization hypothesis selects the boundary value

$$o(n) := \lim_{j \to 1+} q(j) = 2.$$

In this sense, $o(n) = 2$ represents the canonical limiting scale of the two-power decomposition, while $q(j)$ describes the local binomial scale before global normalization.

Under the global normalization hypothesis, this limiting boundary value fixes the scale of the function $o(n)$ uniformly across all exponents:

$$o(n) = 2. \tag{2.15}$$

This normalization locks the base of the growth rate to the binary structure of the equation. Substituting (2.15) into the definition of $o(n)$ yields:

$$2^n = 2n. \tag{2.16}$$

Analysis of the exponential-linear progression $2^n = 2n$. The resulting equation (2.16) represents the direct intersection of the exponential growth 2^n and the linear progression $2n$. In the general case of k decomposed powers, this relation generalizes to the transcendental-like equation $k^n = kn$. For any $k \geq 3$, the exponential progression k^n strictly dominates the linear progression kn for all $n \geq 2$, meaning $n = 1$ is the unique integer root. Only the binary case $k = 2$ possesses the unique algebraic property $2^1 = 2 \cdot 1$ and $2^2 = 2 \cdot 2$, allowing solutions at both $n = 1$ and $n = 2$.

Consider the function $f(n) = (2 \cdot n)^{1/n}$ for $n \geq 1$. Then (2.16) is equivalent to $2 = f(n)$. Note that taking the natural logarithm yields:

$$\ln f(n) = \frac{\ln(2 \cdot n)}{n}, \qquad \frac{d}{dn}\big(\ln f(n)\big) = \frac{1 - \ln(2 \cdot n)}{n^2} < 0 \quad \text{for } n \geq 2,$$

which shows that f is strictly decreasing for $n \geq 2$. The equation $2 = f(n)$ holds exactly for $n \in \{1, 2\}$, since $f(1) = 2$, $f(2) = 2$, and $f(n) < 2$ for all $n > 2$, with $f(n) \to 1$ as $n \to \infty$ (see Appendices B, C).

From this it immediately follows:

- For $n = 1$, we have $2^1 = 2 \cdot 1$, corresponding to trivial linear decompositions.
- For $n = 2$, we have $2^2 = 2 \cdot 2$, corresponding to quadratic decompositions (Pythagorean triples).
- For all $n > 2$, the exponential progression strictly dominates: $2^n > 2n$.

Consequently, the binary scaling constraint $o(n) = 2$ uniquely restricts the allowed exponents to $n \in \{1, 2\}$. For any exponent higher than the square, the balance between the two combined powers on the left-hand side and the linear growth on the right-hand side is broken, which excludes any integer solutions. Substituting (2.15) into the definition of $o(n)$ gives

$$2^n = 2 \cdot n, \tag{2.17}$$

that is

$$n = 2^{n-1}. \tag{2.18}$$

The function $g(n) = n - 2^{n-1}$ is zero exactly at $n = 1$ and $n = 2$, and is negative for all integers $n > 2$ (since the exponential function 2^{n-1} grows faster than the linear function n). Consequently, (2.18) has exactly two positive integer roots: $n = 1$ and $n = 2$.

The alternative form

$$n = \tfrac{1}{2} o(n)^n$$

has a significant advantage: it clearly exposes the asymmetry between linear and exponential growth.

- On the **left-hand side**, we have a linear dependence on n;
- On the **right-hand side**, we have an exponential function $(1/2)o(n)^n$, which grows faster when $o(n) > 1$.

From this representation, several facts become immediately visible:

(a) For integer n, if the equality holds, then $o(n)^n = 2n$, which implies $o(n)^n$ is even.

(b) For the special case of the binary decomposition constraint $o(n) = 2$, we get $n = 2^{n-1}$. The function 2^{n-1} grows faster than n, so the equality is possible only for $n = 1$ and $n = 2$.

(c) Therefore, $o(n)^n$ cannot exceed 4 without violating the balance between the linear left-hand side and the exponentially increasing right-hand side.

Thus, the form $n = \frac{1}{2} o(n)^n$ effectively highlights the core of the argument: the intersection of a linear and an exponential function can occur only at the very beginning, specifically at $n = 1$ and $n = 2$.

Conclusion (in the spirit of Fermat). The structurally justified binary scaling (the two-power decomposition constraint) is a direct manifestation of the modular roots-of-unity filter for $k = 2$. While the local scaling parameter $q(j)$ varies inside the working domain, its symmetric boundary normalization as $j \to 1^+$ provides the canonical limiting value used for the global binary normalization $o(n) = 2$. This binary scaling reduces the transcendental relation $k^n = kn$ specifically to the equation $2^n = 2n$, which uniquely restricts the permissible exponents to $n \in \{1, 2\}$. Consequently, for "exponents higher than the square" there are no solutions — precisely the kind of short, algebraically profound "marginal note" that Fermat himself could have left.

6.) Verification (without considering the total multiplier r from (2.7), which has been reduced)

(a) Consider the case $n = 1$

$$\begin{aligned}
&z = m + p \\
&x = m - p \\
&y = 2 \cdot [C_1^1 (m^1)^{1-(2\cdot 0+1)} (p^1)^{(2\cdot 0+1)} = 2[1 \cdot 1 \cdot p] = 2p
\end{aligned}$$

that is, for the case $n = 1$, we have a solution in natural numbers x, y, z, if m and p - are positive integer numbers and $m > p$.

(b) Consider the case $n = 2$

$$\begin{aligned}
&z = m^2 + p^2 \\
&x = m^2 - p^2 \\
&y = \sqrt{2} \cdot [C_1^2 (m^2)^{2-(2\cdot 0+1)} (p^2)^{(2\cdot 0+1)}]^{1/2} = \sqrt{2} \cdot [2m^2p^2]^{1/2} = 2mp
\end{aligned}$$

that is, for the case $n = 2$, we also have a solution in positive integer numbers x, y, z, if m and p are positive integer numbers and $m > p$, or according to [10], if $m > p$, and m, p are such non-integer numbers that combinations with them will give the positive integer numbers x, y, z, for example: $m = 3/\sqrt{2}$, $p = 1/\sqrt{2}$, in this case, we also get the classical Pythagorean triple:

$$\begin{aligned}
&x = m^2 - p^2 = \left(\tfrac{3}{\sqrt{2}}\right)^2 - \left(\tfrac{1}{\sqrt{2}}\right)^2 = \tfrac{9}{2} - \tfrac{1}{2} = \tfrac{8}{2} = 4 \\
&y = 2 \cdot m \cdot p = 2 \cdot \tfrac{3}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}} = 3 \\
&z = m^2 + p^2 = \left(\tfrac{3}{\sqrt{2}}\right)^2 + \left(\tfrac{1}{\sqrt{2}}\right)^2 = \tfrac{9}{2} + \tfrac{1}{2} = 5
\end{aligned}$$

We have obtained the classical Pythagorean triple and can see that this formula works.

7). Let us now consider the more general case where r is not reduced to proceed to final formulas like (2.7). If r is a positive integer number, then all the above applies. But according to [10], r – can be a rational number. For example, let's say $r = 0.5$:

(a) Consider the case of $n = 1$, $m = 4$, and $p = 2$
$x = 0.5 \cdot (m - p) = 0.5 \cdot (4 - 2) = 1$
$z = 0.5 \cdot (m + p) = 0.5 \cdot (4 + 2) = 3$
$y = 0.5 \cdot (2 \cdot p) = 0.5 \cdot (2 \cdot 2) = 2$
We see positive integer numbers. That is,, m, p must be such that 7*(a)* is performed similarly. In this case, we can see that the formula 6*(a)* works.

(b) Consider the case *of* $n = 2$, $m = 2 \cdot \sqrt{2}$, $p = \sqrt{2}$,
$x = 0.5 \cdot (m^2 - p^2) = 0.5 \cdot \left(\left(2 \cdot \sqrt{2}\right)^2 - \left(\sqrt{2}\right)^2\right) = 3$
$z = 0.5 \cdot (m^2 + p^2) = 0.5 \cdot \left(\left(2 \cdot \sqrt{2}\right)^2 + \left(\sqrt{2}\right)^2\right) = 5$
$y = 0.5 \cdot (2 \cdot m \cdot p) = 0.5 \cdot \left(2 \cdot \left(2 \cdot \sqrt{2}\right) \cdot \left(\sqrt{2}\right)\right) = 4$
We see positive integer numbers. That is,, m, p must be such that 7*(b)* holds similarly. In this case, we can see that the formula 6*(b)* works.

This preserves the generality of solutions for $n = 2$, as seen in the Pythagorean triple example.

8). Therefore, the cases $n = 1$ and $n = 2$ **exhaust all possible integer solutions**, which is consistent with the theorem.
The test showed that for *n=1* or for *n=2*, in all the cases considered, we have solutions of the equation $x^n + y^n = z^n$in the positive integer numbers x, y, z

9). **the equation $x^n + y^n = z^n$ has roots in the positive integer numbers $\boldsymbol{x}$, $\boldsymbol{y}$, $\boldsymbol{z}$ only for**
$\boldsymbol{n} = \mathbf{1}$ and for $\boldsymbol{n} = \mathbf{2}$

This completes the conditional argument
under the global binary-normalization hypothesis.

3 Remark and corollaries

REMARK. *Note that the expression (2.8) can be simplified, namely*

$$\begin{aligned} y &= \sqrt[n]{2}\left[\sum_{i=0}^{k} C_n^{(2i+1)} (m^n)^{n-(2i+1)} (p^n)^{(2i+1)}\right]^{1/n} = \\ &= \sqrt[n]{2}\left[\sum_{i=0}^{k} C_n^{(2i+1)} \frac{(m^n)^n}{(m^n)^{2i} m^n} (p^n)^{2i} p^n\right]^{1/n} = \\ &= \sqrt[n]{2} m^n \frac{1}{m} p \left[\sum_{i=0}^{k} C_n^{(2i+1)} \left(\frac{p}{m}\right)^{n2i}\right]^{1/n} = \\ &= \sqrt[n]{2} m^{n-1} p \left[\sum_{i=0}^{k} C_n^{(2i+1)} \left(\frac{p}{m}\right)^{2in}\right]^{1/n} \end{aligned} \tag{3.1}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even.
COROLLARY 1. *Consider the case $m = p$, then from the expression (3.1) it can be derived that*

$$\begin{aligned} &x = 0 \\ &z = m^n + m^n = 2m^n \\ &y = \sqrt[n]{2} m^{n-1} m \left[\sum_{i=0}^{k} C_n^{(2i+1)} \left(\frac{m}{m}\right)^{2in}\right]^{1/n} = \sqrt[n]{2} m^n \left[\sum_{i=0}^{k} C_n^{(2i+1)}\right]^{1/n} \end{aligned}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even.
It is obvious that $y = z$

$$\begin{aligned} \sqrt[n]{2} m^n \left[\sum_{i=0}^{k} C_n^{(2i+1)}\right]^{1/n} &= 2m^n \\ \sqrt[n]{2} \left[\sum_{i=0}^{k} C_n^{(2i+1)}\right]^{1/n} &= 2 \end{aligned},$$

whence

$$\sum_{i=0}^{k} C_n^{(2i+1)} = 2^{n-1} \tag{3.2}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even.
COROLLARY 2. *Based on (3.2), the sum of even combinations can be calculated.*
Consider Pascal's triangle [7]:

$$\begin{array}{ccccccccccc} & & & & 1 & & 1 & & & & \\ & & & 1 & & 2 & & 1 & & & \\ & & 1 & & 3 & & 3 & & 1 & & \\ & 1 & & 4 & & 6 & & 4 & & 1 & \\ 1 & & 5 & & 10 & & 10 & & 5 & & 1 \end{array}$$

Similarly to the above, it is concluded that

$$\sum_{j=0}^{s} C_n^{2i} = 2^{n-1} \tag{3.3}$$

with $k = (n-1)/2$ if n is odd and $k = (n-2)/2$ if n is even.

PROOF.
Expand

$$(m^n + p^n)^n + (m^n - p^n)^n =$$

into a binomial [8, 9]:

$$\begin{aligned}
&= [(m^n)^n + C_n^1 (m^n)^{n-1} p^n + C_n^2 (m^n)^{n-2} (p^n)^2 + \cdots + C_n^{n-1} m^n (p^n)^{n-1} + (p^n)^n] + \\
&+ [(m^n)^n - C_n^1 (m^n)^{n-1} p^n + C_n^2 (m^n)^{n-2} (p^n)^2 \pm \cdots \pm C_n^{n-1} m^n (p^n)^{n-1} \pm (p^n)^n] = \\
&= 2C_n^0 (m^n)^n + 2C_n^2 (m^n)^{n-2} (p^n)^2 + \ldots + 2C_n^k (m^n)^{n-k} (p^n)^k + \{2C_n^n (p^n)^n\} = \\
&= 2 \sum_{j=0}^{s} C_n^{2j} (m^n)^{n-2j} (p^n)^{2j}
\end{aligned}$$

with $s = (n-1)/2$ if n is odd and $s = n/2$ if n is even.
If $m = p$

$$(p^n + p^n)^n + (p^n - p^n)^n = 2 \sum_{j=0}^{s} C_n^{2j} (p^n)^{n-2j} (p^n)^{2j}$$

$$(2p^n)^n = 2 \sum_{j=0}^{s} C_n^{2j} \frac{(p^n)^n}{(p^n)^{2j}} (p^n)^{2j}$$

$$2^n (p^n)^n = 2(p^n)^n \sum_{j=0}^{s} C_n^{2j}$$

$$2^{n-1} = \sum_{j=0}^{s} C_n^{2j}$$

with $s = (n-1)/2$ if n is odd and $s = n/2$ if n is even.
Corollary 2 is proved.

COROLLARY 3. *Analysing (3.2) and (3.3), it can be concluded that*

$$\sum_{i=0}^{k} C_n^{(2i+1)} = \sum_{j=0}^{s} C_n^{2j} \tag{3.4}$$

with $k = (n-1)/2$, $s = (n-1)/2$ if n is odd and $k = (n-2)/2$, $s = n/2$ if n is even.
Why such borders?

Type of number	**Condition**	**Last constant in equation**
Odd	*k = 2((n-1)/2)+1 = n*	C_n^n
	s = 2((n-1)/2) = n-1	C_n^{n-1}
Even	*k = 2((n-2)/2)+1 = n-1*	C_n^{n-1}
	s = 2(n/2) = n	C_n^n

Table 1: Boundaries of binomial coefficients

Conclusion: the sum of even coefficients is equal to the sum of odd ones and is equal to 2^{n-1}, therefore from (3.4) we have

$$\sum_{r=0}^{n} C_n^r = \sum_{i=0}^{k} C_n^{(2i+1)} + \sum_{j=0}^{s} C_n^{2j} = 2 \cdot 2^{n-1} = 2^n \tag{3.5}$$

with $k = (n-1)/2$, $s = (n-1)/2$ if n is odd and $k = (n-2)/2$, $s = n/2$ if n is even.

4 Conclusion

The "difficulties" were for Fermat the lengthiness of the run of his deductions ***put in writing***, as in the first half of the seventeenth century the mathematical notations had been way far from their present concise and diverse shape, many actions had to be written down ***in words. Besides, a purely mathematical challenge was that he had to operate the then entirely new notions of binomials and logarithms***, both having just appeared for use and to be learnt "on the fly".

As mentioned in the introduction, the mathematical methods from Pierre Fermat's era used in this article's proof are accessible to any first-year physics and mathematics student. This contrasts favorably with Andrew Wiles's proof, which is quite complex for the average mathematician due to its use of advanced and intricate modern mathematical tools.

Fermat was obviously "playing" with the new notions, decomposing powers of differences into sums of powers and suddenly found out that as one confines oneself with positive integers in the power, the logarithmic equation yields immediately that $x^n + y^n = z^n$(which is a difference rewritten as a sum) is correct for whole x, y, z only and if only $n = 1$ or 2.

He (would have) had first to introduce the two new notions so as to fully explain his finding. One can imagine ***how much room it would take to put down all the deliberations*** that had led him to his discovery ***on the margins of a book solely without the proper symbolic notations that a contemporary mathematician avails***.

Why Pierre Fermat did not write down all those ideas in a dedicated document is the dedicated question of a dedicated research endeavour. It can come out that he had authored such a separate document indeed, which afterwards was somehow lost or – alternatively – has survived to this day, hidden in an archive or a library or in somebody's unrealised custody.

The author requests the mathematical society to look critically at the deliberations set forth above and to return their assessment.

This paper has been published as a preprint on the ResearchGate platform and on the OSF platform at the following links [11, 12].

The Coq formalization is available in the repository FLT-Coq on GitHub [14].

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A Appendix A. Proof of the Irrationality of $\sqrt[n]{2}$ for $n \geq 2$

A.1 Theorem: The number $\sqrt[n]{2}$ is irrational for any integer $n \geq 2$.

A.2 Proof:

We will prove by contradiction. Assume that $\sqrt[n]{2}$ is a rational number. This means it can be represented as an irreducible fraction $\frac{p}{q}$, where p and q are integers, $q \neq 0$, and p and q have no common divisors other than 1.

Thus, we have:

$$\sqrt[n]{2} = \frac{p}{q}$$

Raise both sides of the equation to the power of n:

$$(\sqrt[n]{2})^n = \left(\frac{p}{q}\right)^n$$

$$2 = \frac{p^n}{q^n}$$

Multiply both sides by q^n:

$$2q^n = p^n$$

From this equation, we see that p^n is an even number, since it is equal to $2q^n$. If p^n is even, then p must also be even (if p were odd, then p^n would also be odd).

Since p is even, we can write it as $p = 2k$, where k is some integer. Substitute this expression for p into the equation $2q^n = p^n$:

$$2q^n = (2k)^n$$

$$2q^n = 2^n k^n$$

Divide both sides of the equation by 2 (which is permissible since $n \geq 2$, and therefore 2^n is divisible by 2):

$$q^n = 2^{n-1}k^n$$

Since $n \geq 2$, then $n - 1 \geq 1$. From the last equation, we see that q^n is an even number, since it is equal to $2^{n-1}k^n$, where 2^{n-1} is an even factor. If q^n is even, then q must also be even.

Thus, we have reached the conclusion that both p and q are even numbers. This means they have a common divisor of 2. However, at the beginning of the proof, we assumed that the fraction $\frac{p}{q}$ was irreducible, meaning p and q have no common divisors other than 1.

The resulting contradiction indicates that our initial assumption about the rationality of $\sqrt[n]{2}$ is incorrect.

A.3 Conclusion:

Therefore, the number $\sqrt[n]{2}$ is irrational for any integer $n \geq 2$.

B Appendix B. Proof that the limit of the function $f(n) = \sqrt[n]{2 \cdot n}$ is 1

B.1 Problem Statement

Prove that the limit of the function $f(n) = \sqrt[n]{2 \cdot n}$ as n approaches infinity is equal to 1.

B.2 Solution

Consider the function $f(n) = \sqrt[n]{2 \cdot n}$. Our goal is to find the limit of this function as $n \to \infty$. Let's rewrite the function in the form of a power:

$$f(n) = (2 \cdot n)^{\frac{1}{n}}$$

To find the limit of this function, let's consider the natural logarithm of $f(n)$:

$$\ln(f(n)) = \ln\left((2 \cdot n)^{\frac{1}{n}}\right)$$

Using the properties of logarithms, we get:

$$\ln(f(n)) = \frac{1}{n}\ln(2 \cdot n)$$

Separate the logarithm of the product into the sum of logarithms:

$$\ln(f(n)) = \frac{\ln(2) + \ln(n)}{n}$$

Now, let's find the limit of $\ln(f(n))$ as $n \to \infty$:

$$\lim_{n\to\infty} \ln(f(n)) = \lim_{n\to\infty} \frac{\ln(2) + \ln(n)}{n}$$

This limit has the indeterminate form $\frac{\infty}{\infty}$, so we can apply L'Hôpital's Rule. Take the derivatives of the numerator and denominator with respect to n:

$$\frac{d}{dn}(\ln(2) + \ln(n)) = \frac{1}{n}$$

$$\frac{d}{dn}(n) = 1$$

Thus, the limit becomes:

$$\lim_{n\to\infty} \frac{\frac{1}{n}}{1} = \lim_{n\to\infty} \frac{1}{n} = 0$$

So, we have found that the limit of the natural logarithm of the function is 0:

$$\lim_{n\to\infty} \ln(f(n)) = 0$$

Now, to find the limit of the original function $f(n)$, we use the continuity of the exponential function:

$$\lim_{n\to\infty} f(n) = \lim_{n\to\infty} e^{\ln(f(n))} = e^{\lim_{n\to\infty} \ln(f(n))} = e^0 = 1$$

B.3 Conclusion

Therefore, we have proven that the limit of the function $f(n) = \sqrt[n]{2 \cdot n}$ as n approaches infinity is equal to 1.

C Appendix C. A short proof of monotonicity

It is well known that $\sqrt[n]{2 \cdot n} \to 1$ as $n \to \infty$ (see Appendix B). However, in this appendix we focus on showing that $(2 \cdot n)^{1/n}$ is strictly decreasing for $n \geq 3$. Below is one way to prove this, using the logarithmic derivative:

C.1 Define the function.

Let
$$f(n) \;=\; (2 \cdot n)^{\frac{1}{n}}.$$
It is convenient to work with its natural logarithm:
$$\ln\bigl(f(n)\bigr) \;=\; \ln\bigl((2 \cdot n)^{\frac{1}{n}}\bigr) \;=\; \frac{1}{n}\,\ln(2 \cdot n).$$
Denote
$$g(n) \;=\; \ln\bigl(f(n)\bigr) \;=\; \frac{\ln(2 \cdot n)}{n} \;=\; \frac{\ln(2) + \ln(n)}{n}.$$

C.2 Compute the derivative.

Treating n as a real variable $n > 0$, we have
$$g'(n) \;=\; \frac{d}{dn}\Bigl(\tfrac{\ln(2 \cdot n)}{n}\Bigr).$$
Using the quotient rule,
$$g'(n) = \frac{\frac{d}{dn}[\ln(2 \cdot n)] \cdot n \;-\; \ln(2 \cdot n) \cdot 1}{n^2} = \frac{\frac{1}{n} \cdot n \;-\; \ln(2 \cdot n)}{n^2} = \frac{1 - \ln(2 \cdot n)}{n^2}.$$
(Note that $\frac{d}{dn}[\ln(2 \cdot n)] = \frac{1}{2 \cdot n} \cdot 2 = \frac{1}{n}$.)

C.3 Sign of the derivative.

We want to see where $g'(n) < 0$:
$$g'(n) < 0 \iff 1 - \ln(2 \cdot n) < 0 \iff \ln(2 \cdot n) > 1 \iff 2 \cdot n > e \iff n > \frac{e}{2}.$$

Since $e \approx 2.718$, the inequality $n > e/2$ is certainly true for all integer $n \geq 2$, and hence strictly for $n \geq 3$. Therefore, for $n \geq 3$, $g(n)$ is strictly decreasing.

C.4 Implication for $f(n)$.

Since $f(n) = \exp(g(n))$, and $\exp(\cdot)$ is a strictly increasing function in its argument, $f(n)$ decreases whenever $g(n)$ decreases. Hence, $f(n)$ is indeed strictly decreasing for $n \geq 3$.

Thus, only for $n = 1$ and $n = 2$ does $f(n)$ attain the maximum value 2, and for all $n \geq 3$, it strictly decreases (which is consistent with the limit $\lim_{n\to\infty}(2 \cdot n)^{1/n} = 1$).

C.5 Conclusion

Hence, we have demonstrated that $f(n)$ is strictly decreasing for $n \geq 3$, either by analyzing the derivative of $\ln(f(n))$ (as shown above) or by comparing $f(n+1)$ and $f(n)$. This completes the proof of the monotonicity of $f(n)$.

D Appendix D: Analysis of Functions and Heuristic Visualization of Critical Points

D.1 Introduction

The goal of this analysis is to study the behavior of the auxiliary continuous function

$$f(p, q) = q - \sqrt[p]{qp},$$

and to identify its critical points, in particular the points where the second derivative $f''(p)$ equals zero. This study is not intended as an independent proof of Fermat's Last Theorem. Rather, it provides a geometric and analytic visualization of the special role of the binary normalization value $o = 2$ in the context of the conditional scaling scheme discussed in the main article:

$$x^n + y^n = z^n.$$

In this appendix, the symbol q denotes an auxiliary real parameter in the function $f(p, q)$. It should not be confused with a modulus q used in modular arithmetic or with the finite field notation $\mathbb{F}_q$. In the present geometric visualization, the number o represents the global scaling parameter associated with the binary symmetry of the equation, and the auxiliary value $q = 2$ corresponds to the same normalized binary scale. The special role of $q = 2$, or equivalently $o = 2$, is therefore interpreted as a geometric illustration of the algebraic binary-normalization hypothesis used in the main text.

D.2 Results of the Analysis

D.2.1 Combined Graph of Functions

The following graph (Fig. 1) illustrates three functions, each with its corresponding discrete values:

- $1.5 - \sqrt[p]{1.5p}$ (blue line) and $1.5 - \sqrt[n]{1.5n}$ (blue squares),
- $2 - \sqrt[p]{2p}$ (green line) and $2 - \sqrt[n]{2n}$ (green squares),
- $3 - \sqrt[p]{3p}$ (orange line) and $3 - \sqrt[n]{3n}$ (orange squares).

These curves should be read as a visual comparison between continuous values of p and the discrete exponent values n. The case $q = 2$ is singled out because it corresponds to the binary scaling value used in the normalized relation

$$2^n = 2n.$$

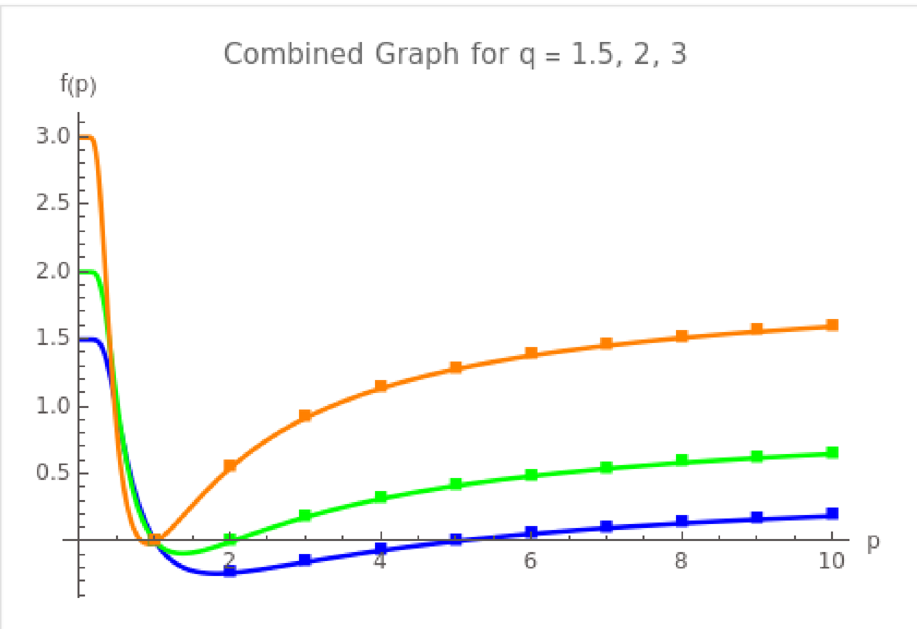

Figure 1: Combined graph of $q - \sqrt[p]{qp}$ for $q = 1.5, 2, 3$. Lines represent continuous p, and squares represent discrete n.

D.2.2 Surface Plot of $f(p,q)$

The three-dimensional surface plot below (Fig. 2) illustrates the behavior of

$$f(p,q) = q - \sqrt[p]{qp}$$

for continuous values of $p > 0$ and $q > 0$. The perspective highlights the change in curvature as both parameters vary. This visualization is meant to support the geometric intuition behind the binary value $q = 2$, not to replace the algebraic and Coq-verified conditional argument given in the main part of the paper.

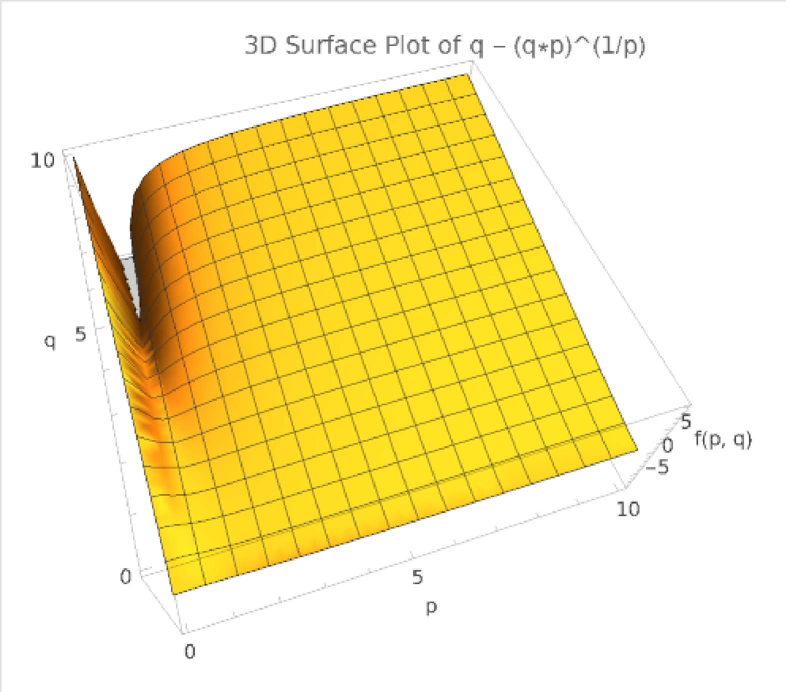

Figure 2: 3D surface plot of $f(p,q) = q - \sqrt[p]{qp}$ for $p > 0$ and $q > 0$.

D.3 Analysis and Conclusions

The numerical and graphical analysis suggests the following interpretation:

- The critical point at $p = 2$ appears naturally for $q = 2$, where the second derivative $f''(p)$ equals zero. This highlights the special geometric role of $q = 2$, corresponding to the binary normalization value $o = 2$.
- For other values of q, critical points may exist, but they occur at values $p \neq 2$. This illustrates that $q = 2$ is distinguished by its alignment with the quadratic case $n = 2$, which is precisely the case of Pythagorean triples.

These results should be understood as a geometric consistency check for the algebraic argument of the article. They illustrate the same special role of the binary value

$$o = 2,$$

but they do not constitute an independent proof of Fermat's Last Theorem. The rigorous conditional part of the argument remains the algebraic reduction to

$$2^n = 2n$$

under the global binary-normalization hypothesis, together with the Coq-verified fact that this equation has only the positive integer solutions

$$n \in \{1, 2\}.$$

E Appendix E: Geometric Motivation for the Binary Normalization through the Function $f(p, q)$

In this appendix, we further investigate the connection between the auxiliary function

$$f(p, q) = q - \sqrt[p]{qp}$$

and the normalized scaling relation associated with Fermat's equation

$$x^n + y^n = z^n.$$

The purpose of this appendix is not to provide a standalone geometric proof of Fermat's Last Theorem, but to visualize why the value $q = 2$, corresponding to the global scaling choice $o = 2$, is geometrically distinguished.

In the conditional scheme of the main article, the decisive algebraic step is not the graph itself, but the normalized relation

$$o(n)^n = 2n$$

together with the binary-normalization hypothesis

$$o(n) = 2.$$

The function $f(p, q)$ gives a visual way to see why the point $p = 2,\ q = 2$ plays a special role: it marks the place where the binary value aligns with the quadratic case, that is, with the existence of Pythagorean triples.

E.1 Analysis of the Second Derivative

The function

$$f(p, q) = q - \sqrt[p]{qp}$$

allows us to study the behavior of inflection points, which are determined by the condition

$$f''_p(p, q) = \frac{\partial^2}{\partial p^2}\left(q - \sqrt[p]{qp}\right) = 0.$$

Inflection points are useful here because they reveal changes in the shape of the auxiliary function. In the present context, these changes are interpreted as a geometric reflection of the transition from the quadratic case $n = 2$ to higher exponents $n > 2$.

E.2 Additional Analysis for the Variable q

Investigations were also conducted on the variable q. At the point $q = 2$, with $p = 2$, both the first and second partial derivatives with respect to q are equal to 0.5. This result indicates a special local behavior of the auxiliary function at the binary point.

However, in the main argument of the article, the value $q = 2$ is not introduced as a consequence of this numerical observation. Rather, it is selected by the global binary-normalization hypothesis, motivated by the roots-of-unity filter and by the boundary normalization

$$\lim_{j \to 1+} q(j) = 2.$$

Thus, the differential analysis in this appendix should be read as an additional geometric illustration of the same binary value, not as a replacement for the algebraic normalization argument.

E.3 Numerical Results

Below are the numerical values of the second derivative $f''_p(p,q)$ for various values of p and q:

p	$q = 1.5$	$q = 2.0$	$q = 2.5$	$q = 3.0$	$q = 4.0$
1	2.75	0.77	2.90	0.96	1.10
2	0.17	0.00	-0.11	-0.34	-0.55
3	-0.0056	-0.08	-0.099	-0.21	-0.31
4	-0.018	-0.11	-0.056	-0.12	-0.17
5	-0.014	-0.10	-0.033	-0.067	-0.093

Table 2: Numerical values of the second derivative $f''_p(p,q)$ for various p and q.

The table shows that the pair

$$p = 2, \qquad q = 2$$

is singled out by the vanishing of the second derivative in the numerical data. This is consistent with the special role of the quadratic case, but it should not be interpreted as a direct proof that higher-degree integer solutions are impossible.

E.4 Conclusions from the Data

- For $p = 2,\ q = 2$ we have $f''_p(p,q) = 0$, which geometrically corresponds to the quadratic case and is consistent with the existence of Pythagorean triples.
- For $q > 2$, the inflection point shifts toward lower values of p. This suggests a change in the shape of the auxiliary function and is consistent with the algebraic exclusion of higher exponents under the binary-normalization hypothesis.
- Figure 3 illustrates the surface of $f''_p(p,q)$ with the marked inflection points.

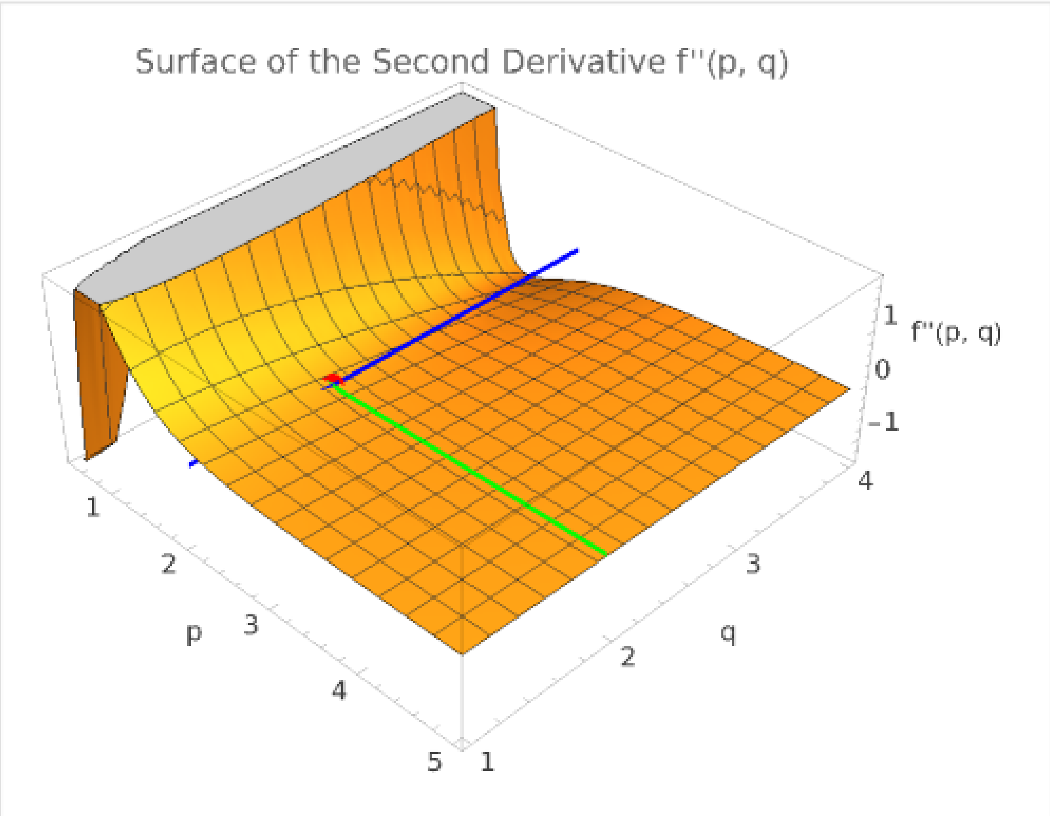

Figure 3: Surface of the second derivative $f''_p(p,q)$, demonstrating the shift of the inflection point.

E.5 Geometric Interpretation in Relation to Fermat's Equation

E.5.1 Relation to the Inflection Points

From the data, one can draw the following heuristic interpretation:

1. If the inflection point is aligned with the discrete exponent value $p = n$, then the auxiliary geometry is compatible with the corresponding exponent.
2. The alignment $p = 2,\ q = 2$ is consistent with the quadratic case $n = 2$, where integer solutions exist in the form of Pythagorean triples.
3. For $q > 2$, the shift of the inflection point away from $p = 2$ illustrates the loss of this quadratic alignment.

Thus, the geometric picture distinguishes the quadratic case

$$n = 2,$$

while the exclusion of exponents

$$n > 2$$

comes from the algebraic and Coq-verified conditional relation

$$2^n = 2n \quad \Longrightarrow \quad n \in \{1, 2\}.$$

E.5.2 Final Conclusion

Based on the numerical and graphical calculations:

- **For $q = 2$, the inflection point corresponds to $p = 2$** $\Rightarrow$ the auxiliary geometry is aligned with the quadratic case, where Fermat's equation has integer solutions through Pythagorean triples.
- **For $q > 2$, the inflection point shifts to lower values of p** $\Rightarrow$ the auxiliary geometry no longer has the same alignment with the quadratic case.
- This supports the interpretation that the value

  $$o = 2$$

  is the natural binary normalization value in the conditional framework of the article.

Thus, the function $f(p, q)$ provides a useful geometric visualization of the special role of the binary value $q = 2$, or equivalently $o = 2$. It supports and illustrates the main conditional algebraic argument, but the strict exclusion of integer solutions for

$$x^n + y^n = z^n, \quad n > 2,$$

rests on the normalized relation

$$2^n = 2n$$

and on the separately stated global binary-normalization hypothesis.